Supplementary Material for Residual Corrected Flow for Human-Centric Editing Across Video Shots and 3D Assets

Wenshuo Gao, Jiangyue Zeng, and Shuai Yang, *Member, IEEE,*

This supplementary material provides the full algorithms, evaluation details, additional visual results, and a multi-shot video editing application. Appendix A describes condition preparation and inference for video and 3D editing. Appendix B describes the details of VLM scoring. Appendices C and D present more video and 3D editing results, respectively. Appendix E provides additional qualitative comparisons. Appendix F presents the multi-shot video editing application.

Appendix A
Full Algorithm

This section describes condition preparation and inference for both branches. We write the updates in latent space and omit the internal operations of the pretrained encoders and decoders. Masks are resized or mapped to the corresponding latent coordinates before they are used.

A. Video Conditions

Given a source video X, source and target descriptions $q_{\mathrm{src}}, q_{\mathrm{tar}}$, and an edit instruction p, we prepare appearance and structural conditions separately. FASHN Human Parser initializes person and face/hair masks from the first frame, and MatAnyone propagates the two masks independently, giving the person mask P and head mask H throughout the video.

a) Source and target appearance: The source reference r_{src} is the first video frame. The image editor produces $r_{\mathrm{tar}} = G(r_{\mathrm{src}}, p, \mathcal{A})$, where $\mathcal{A}$ contains any enabled background, clothing, or object references and may be empty. The instruction identifies the role of each supplied image.

b) Shared structural control: Source depth D, Canny edges C, and HED boundaries E form the regional control

$$S = H \odot (w_D D + w_C C + w_E E) + (P - H) \odot D.$$

We use $w_D = 0.8$ and $w_C = w_E = 0.1$. The head therefore receives depth and edge information, the remaining person receives depth, and the background receives no structural constraint. Both branches use the same S, with conditions $c_{\mathrm{src}} = (q_{\mathrm{src}}, r_{\mathrm{src}}, S)$ and $c_{\mathrm{tar}} = (q_{\mathrm{tar}}, r_{\mathrm{tar}}, S)$.

B. Video Inference

We encode X into z_0 and sample one Gaussian noise ϵ for the entire trajectory. The known source path is $z_t = (1-t)z_0 + t\epsilon$, with denoising velocity $V_0 = z_0 - \epsilon$. At each refresh step t_i, the source prediction gives $V^{\mathrm{res}}_{t_i} = V_0 - V^{\mathrm{src}}_{t_i}(z_{t_i})$. Between refreshes, $V^{\mathrm{res}}_{t_i}$ stores the last computed residual, while the target velocity is evaluated at every step. We use a reuse interval of $r = 10$. With one target prediction per step and one source prediction every r steps, this requires only $1 + 1/r = 1.1$ times the flow evaluations of pure generation under the same guidance settings. Applying the residual within H preserves identity-bearing content without imposing the same reconstruction constraint on clothing or background:

$$V^{\mathrm{edit}}_{t_i}(z^{\mathrm{edit}}_{t_i}) = V^{\mathrm{tar}}_{t_i}(z^{\mathrm{edit}}_{t_i}) + H \odot V^{\mathrm{res}}_{t_i}.$$

After each update, source high frequencies are transferred within the head region. Let LF and HF denote the low- and high-frequency components. The transfer operator is

$$\mathcal{T}_H(u, z) = u + \lambda H \odot [\mathrm{HF}(z) - \mathrm{HF}(u)].$$

We use a frequency threshold of 0.8 for the low/high-frequency split and set the source high-frequency weight to $\lambda = 0.5$. Outside H, the operator leaves the edited latent unchanged. The source latent is taken at the same noise level as the updated target. Algorithm S1 gives the complete inference procedure.

C. 3D Conditions

The supplied mesh M is encoded into shape features z_0^{sh} and appearance features z_0^{tex} on a common sparse coordinate set $\mathcal{C}$. Source views supply the appearance condition, and the image editor constructs a target view from the source frontal view, edit instruction, and optional clothing reference. The source and target images use a shared source-derived crop and are separately masked to isolate the person.

a) Spatial head mask: Face and hair labels from the source views are projected onto the observed surface. These labels are associated with the sparse latent coordinates to obtain $H_{\mathcal{C}}$. Both editing stages use this source-derived spatial mask, so preservation is defined on the asset rather than separately in each rendered view.

b) Shape and appearance conditions: The shape stage uses the source and target image conditions to edit geometry. The appearance stage additionally receives shape features: source shape conditions the source branch, while the edited shape conditions the target branch. Thus, material editing follows the edited geometry while source reconstruction remains tied to the supplied asset.

Algorithm S1 Residual-Corrected Flow for Video Editing

Require: Initial noise ϵ, source latent z_0, source and target conditions $c_{\text{src}}, c_{\text{tar}}$, schedule $0 = t_0 < \cdots < t_N = 1$, head mask H, reuse interval r, high-frequency weight λ

Ensure: Edited latent z_0^{edit}

$V_0 \leftarrow z_0 - \epsilon$; $z_{t_N}^{\text{edit}} \leftarrow \epsilon$
for $i = N, N-1, \ldots, 1$ **do**
  $z_{t_i} \leftarrow (1 - t_i) z_0 + t_i \epsilon$
  $z_{t_{i-1}} \leftarrow (1 - t_{i-1}) z_0 + t_{i-1} \epsilon$
  if $(N - i) \bmod r = 0$ **then**
    $V_{t_i}^{\text{src}}(z_{t_i}) \leftarrow F_\theta(z_{t_i}, t_i, c_{\text{src}})$
    $V_{t_i}^{\text{res}} \leftarrow V_0 - V_{t_i}^{\text{src}}(z_{t_i})$
  else
    $V_{t_i}^{\text{res}} \leftarrow V_{t_{i+1}}^{\text{res}}$
  end if
  $V_{t_i}^{\text{tar}}(z_{t_i}^{\text{edit}}) \leftarrow F_\theta(z_{t_i}^{\text{edit}}, t_i, c_{\text{tar}})$
  $V_{t_i}^{\text{edit}} \leftarrow V_{t_i}^{\text{tar}}(z_{t_i}^{\text{edit}}) + H \odot V_{t_i}^{\text{res}}$
  $z_{t_{i-1}}^{\text{edit}} \leftarrow z_{t_i}^{\text{edit}} + (t_i - t_{i-1}) V_{t_i}^{\text{edit}}$
  $z_{t_{i-1}}^{\text{edit}} \leftarrow \text{LF}(z_{t_{i-1}}^{\text{edit}}) + \lambda H \odot \text{HF}(z_{t_{i-1}}) + (1 - \lambda H) \odot \text{HF}(z_{t_{i-1}}^{\text{edit}})$
end for
return z_0^{edit}

D. 3D Inference

Shape and appearance are edited sequentially. Within each stage, z_0 denotes the source representation, and $V_{t_i}^{\text{src}}$ and $V_{t_i}^{\text{tar}}$ denote the stage-specific fields under its source and target conditions. We use the same denoising convention as the main paper: $V_0 = z_0 - \epsilon$, and the update from t_i to t_{i-1} adds $(t_i - t_{i-1}) V_{t_i}$.

The residual $V_{t_i}^{\text{res}} = V_0 - V_{t_i}^{\text{src}}(z_{t_i})$ corrects the source reconstruction error. To apply frequency transfer, we first form the unmasked residual-corrected proposal $\widetilde{z}_{t_{i-1}}^{\text{edit}}$ and its difference Δ_i from the known source latent $z_{t_{i-1}}$. The spatial head mask then blends the filtered correction with the ordinary target step. This reduces to the masked velocity update:

$$V_{t_i}^{\text{edit}}(z_{t_i}^{\text{edit}}) = V_{t_i}^{\text{tar}}(z_{t_i}^{\text{edit}}) + H_{\mathcal{C}} \odot V_{t_i}^{\text{res}}.$$

Frequency transfer uses the actual coordinates $\mathcal{C}$ and the occupancy-normalized operator $\text{LP}_{b,\mathcal{C}}$ defined in the main paper. The transfer operator is

$$\mathcal{T}(\Delta_i) = \text{LP}_{b,\mathcal{C}}(\Delta_i) + \eta[\Delta_i - \text{LP}_{b,\mathcal{C}}(\Delta_i)].$$

We use filtering strength $b = 0.2$ and retained high-frequency fraction $\eta = 0.5$ in both stages. The source prediction is recomputed at each 3D step. Algorithm S2 gives the procedure; the stage variable s selects the shape or appearance model and its corresponding conditions.

Algorithm S2 Residual-Corrected Flow for 3D Editing

Require: Source latents $z_0^{\mathrm{sh}}, z_0^{\mathrm{tex}}$, source and target image conditions, schedule $0 = t_0 < \cdots < t_N = 1$, spatial head mask $H_{\mathcal{C}}$, frequency-transfer operator $\mathcal{T}$
Ensure: Edited latents $z_0^{\mathrm{sh,edit}}, z_0^{\mathrm{tex,edit}}$

for $s \in \{\mathrm{sh}, \mathrm{tex}\}$, in this order **do**
  $z_0 \leftarrow z_0^s$; form stage conditions $c^s_{\mathrm{src}}, c^s_{\mathrm{tar}}$
  if $s = \mathrm{tex}$ **then**
    Use z_0^{sh} in the source branch and $z_0^{\mathrm{sh,edit}}$ in the target branch
  end if
  Sample fixed Gaussian noise ϵ; $V_0 \leftarrow z_0 - \epsilon$; $z_{t_N}^{\mathrm{edit}} \leftarrow \epsilon$
  for $i = N, N-1, \ldots, 1$ **do**
    $z_{t_i} \leftarrow (1 - t_i) z_0 + t_i \epsilon$
    $z_{t_{i-1}} \leftarrow (1 - t_{i-1}) z_0 + t_{i-1} \epsilon$
    $V^{\mathrm{src}}_{t_i}(z_{t_i}) \leftarrow F^s_\theta(z_{t_i}, t_i, c^s_{\mathrm{src}})$
    $V^{\mathrm{tar}}_{t_i}(z^{\mathrm{edit}}_{t_i}) \leftarrow F^s_\theta(z^{\mathrm{edit}}_{t_i}, t_i, c^s_{\mathrm{tar}})$
    $V^{\mathrm{res}}_{t_i} \leftarrow V_0 - V^{\mathrm{src}}_{t_i}(z_{t_i})$
    $\bar{z}^{\mathrm{tar}}_{t_{i-1}} \leftarrow z^{\mathrm{edit}}_{t_i} + (t_i - t_{i-1}) V^{\mathrm{tar}}_{t_i}(z^{\mathrm{edit}}_{t_i})$
    $\tilde{z}^{\mathrm{edit}}_{t_{i-1}} \leftarrow \bar{z}^{\mathrm{tar}}_{t_{i-1}} + (t_i - t_{i-1}) V^{\mathrm{res}}_{t_i}$
    $\Delta_i \leftarrow \tilde{z}^{\mathrm{edit}}_{t_{i-1}} - z_{t_{i-1}}$
    $z^{\mathrm{edit}}_{t_{i-1}} \leftarrow (1 - H_{\mathcal{C}}) \odot \bar{z}^{\mathrm{tar}}_{t_{i-1}} + H_{\mathcal{C}} \odot [z_{t_{i-1}} + \mathcal{T}(\Delta_i)]$
  end for
  $z_0^{s,\mathrm{edit}} \leftarrow z^{\mathrm{edit}}_{t_0}$
end for
return $z_0^{\mathrm{sh,edit}}, z_0^{\mathrm{tex,edit}}$

APPENDIX B
VLM SCORING

We use GPT-6.1-sol to score video edits along three dimensions: Instruction and Reference Following (IR), Video Quality (VQ), and Human Consistency (HC). For each source–output pair, we uniformly sample eight aligned frame pairs, including the first and last frames, and retain the full image area. The input contains the edit instruction, the supplied reference images labeled by role (scene, clothing, or object), and the source and edited frames in chronological order. Only references supplied for that editing case are included. Each method output is evaluated independently with the same prompt. The evaluator returns an integer score from 1 to 5 and a short reason for each criterion. We average the case-level scores separately for HC, IR, and VQ to obtain final result.

The following prompt is shared across all evaluated methods. The case-specific instruction, references, and frame pairs are appended in the indicated order.

> You are evaluating a human-centric video edit. You are given an edit instruction, optional reference images labeled by role, and eight chronologically ordered pairs of source and edited frames. Each pair shows the same time point. Evaluate the edited sequence using the following three criteria independently.
> **Instruction and Reference Following (IR).** Assess whether the requested scene, clothing, object, and lighting changes are achieved. Compare the relevant attributes with each supplied reference image and check whether the requested design remains consistent across frames. For text-only edits, use the edit instruction as the target. Changes explicitly requested by the instruction are desirable.
> **Video Quality (VQ).** Assess the edited sequence's visual quality: natural appearance, clear details, coherent lighting and shadows, plausible anatomy and materials, and clean boundaries. Consider visible artifacts, distortions, and appearance instability across the supplied frames.
> **Human Consistency (HC).** Compare the person in the edited sequence with the aligned source sequence and jointly assess four aspects. Structure: preservation of body proportions, pose, and the spatial relationships among body parts. Detail: preservation of fine facial, hair, and surface details in regions that should remain unchanged. Motion: agreement with the recorded movements, gestures, and expression changes, together with coherent evolution across frames. Identity: preservation of the same person's facial features and appearance throughout the sequence. Give one score that reflects all four aspects. Requested changes to clothing, lighting, or other instructed attributes should not be treated as consistency errors by themselves.
> **Scoring scale.** Use an integer from 1 to 5 for each criterion: 1 = severe failure; 2 = substantial errors; 3 = partially successful with noticeable errors; 4 = successful with minor imperfections; 5 = excellent with negligible errors. Apply these anchors to the requirements of each criterion.
> Return only a JSON object with the keys IR, VQ, and HC. Each key must contain an object with an integer score and a one-sentence reason. Base each reason on specific visual evidence from the supplied frames or references.

APPENDIX C
MORE VIDEO EDITING RESULTS

We present more video editing results in Figs. S1–S3.

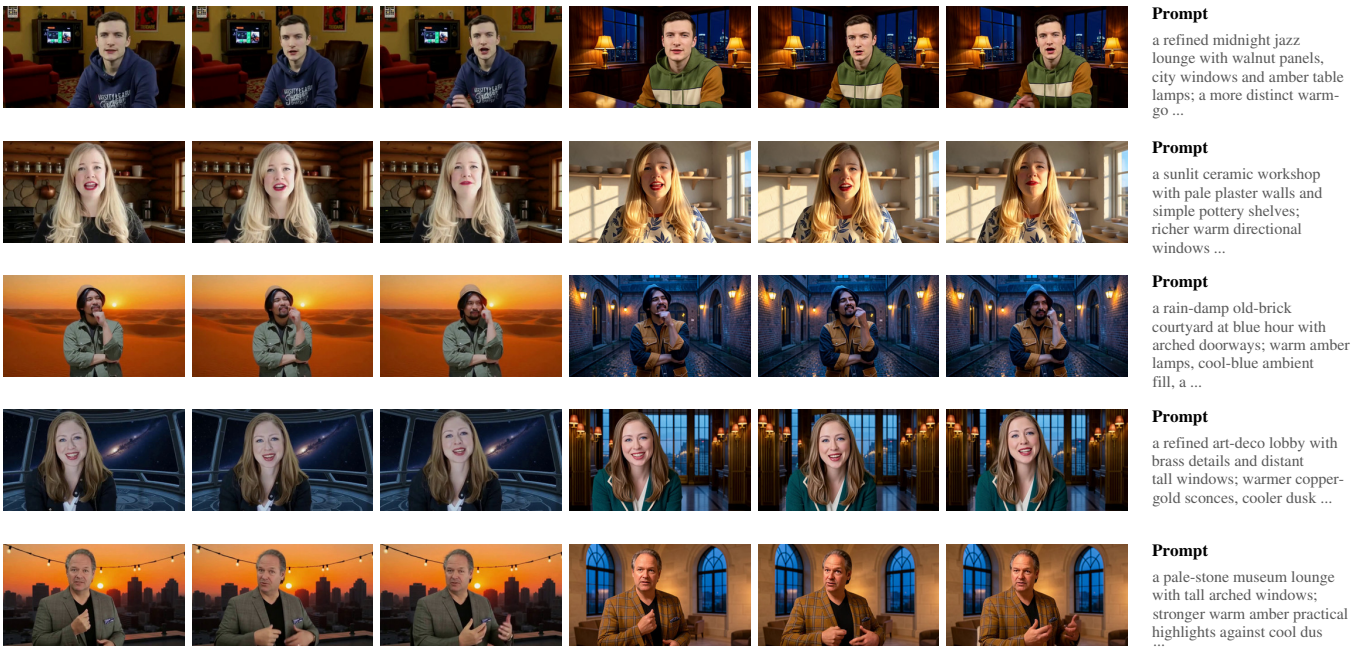


Fig. S1. More text-guided video editing results.

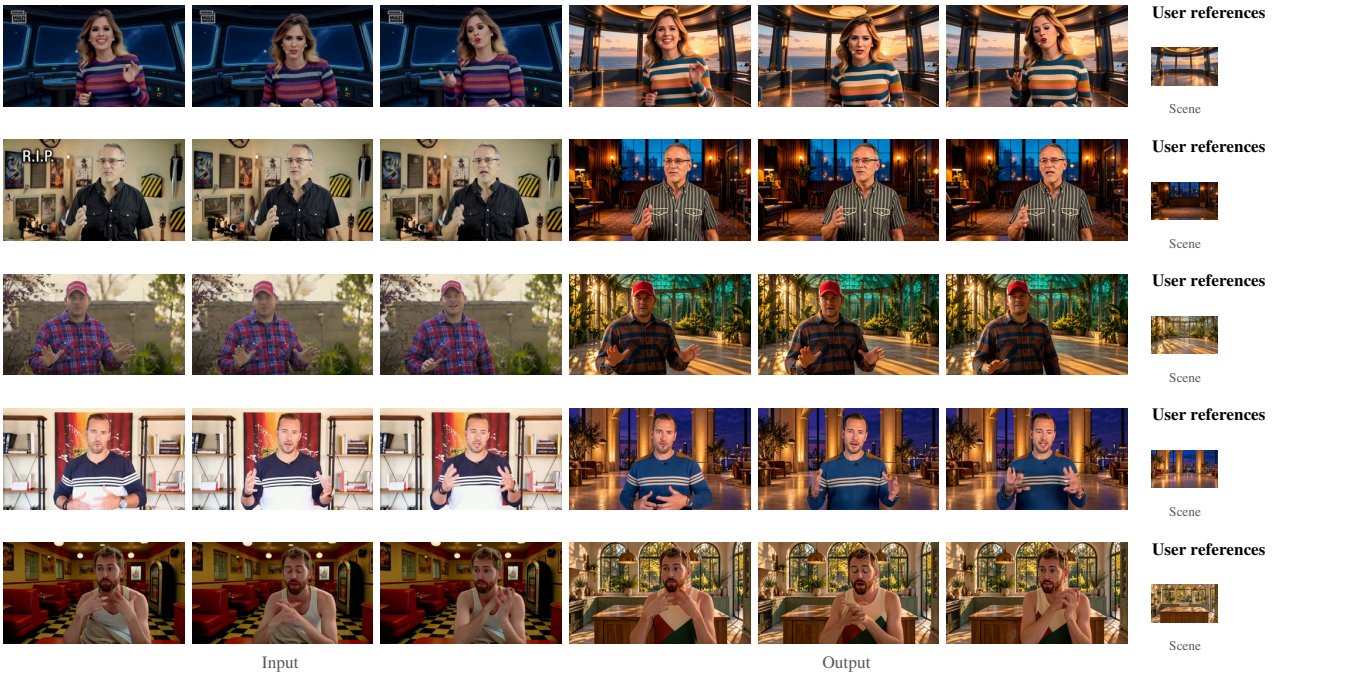


Fig. S2. More background-guided video editing results.

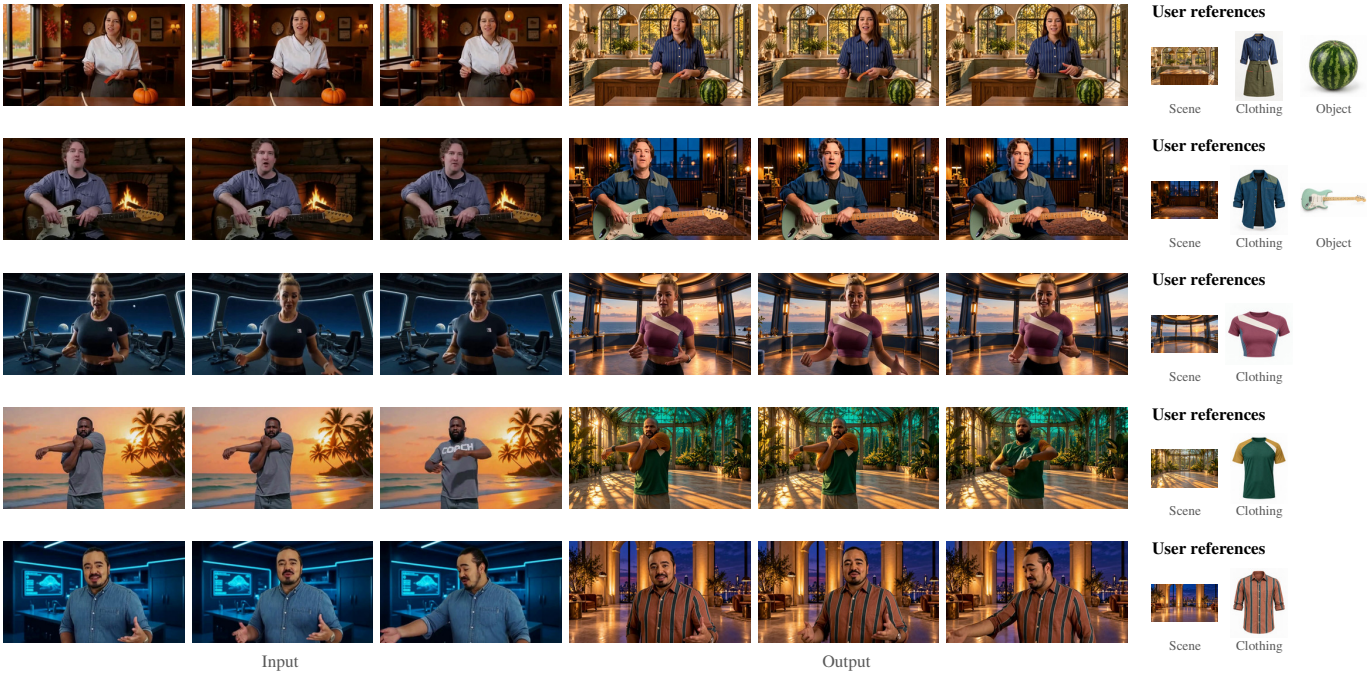


Fig. S3. More reference-guided video editing results.



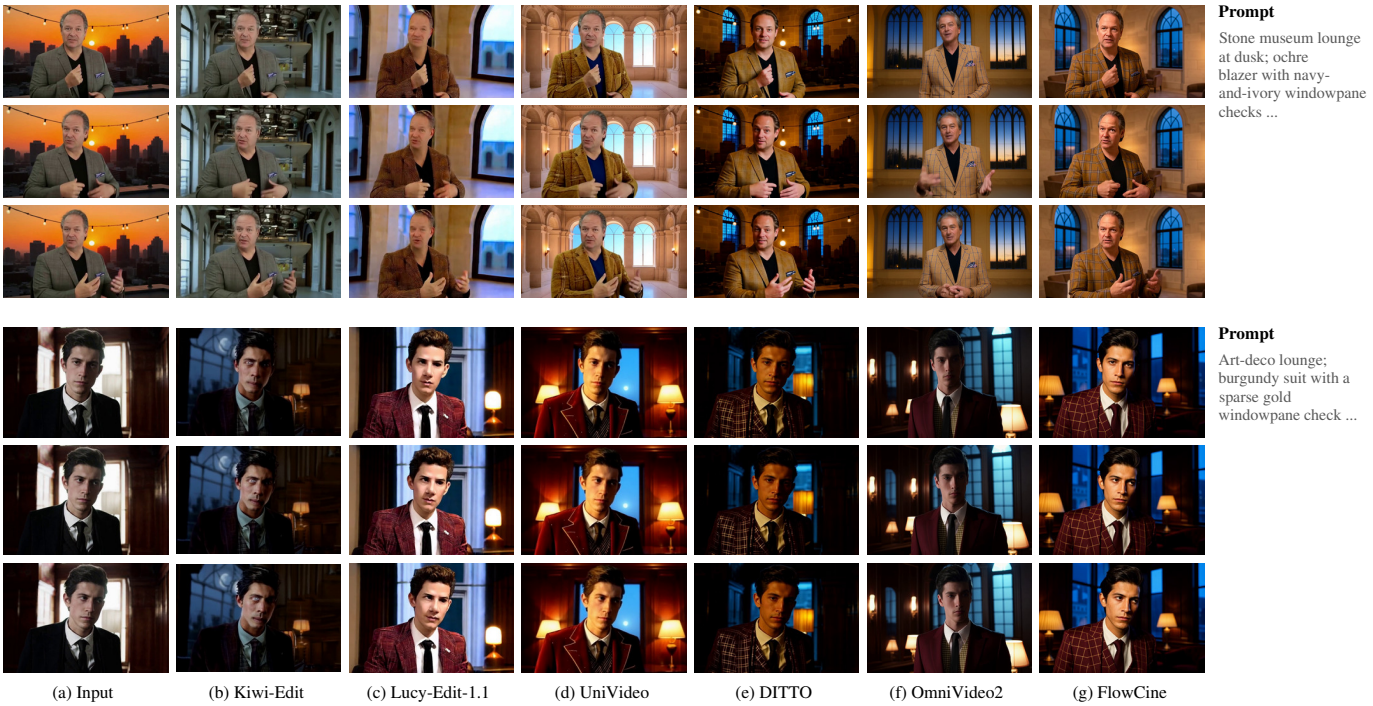
Fig. S4. More 3D editing results.

APPENDIX D
MORE 3D EDITING RESULTS

We present more 3D editing results in Fig. S4.

APPENDIX E
MORE COMPARISONS

We present more qualitative video comparisons in Fig. S5.



(a) Input (b) Kiwi-Edit (c) Lucy-Edit-1.1 (d) UniVideo (e) DITTO (f) OmniVideo2 (g) FlowCine

Fig. S5. More qualitative video comparisons.



Fig. S6. Multi-shot video editing with scene, clothing, and object references. Source and edited frames are shown from distinct shots.

APPENDIX F
APPLICATION: MULTI-SHOT VIDEO EDITING

A coherent visual design is essential when edited shots are assembled into a sequence. Changes to the setting, clothing, and props should remain recognizable across camera angles and compositions, while each shot retains the actor's identity and recorded performance. FlowCine supports this workflow by extending its editing interface from individual shots to a shared design across shots.

Users specify a common edit instruction and may supply background, clothing, and object references independently. Each shot generates its own target appearance from these shared conditions and its source frame, adapting the design to the local pose and composition. Region-dependent structural control follows the recorded performance, while head-focused residual correction and detail transfer preserve identity-related features. Shots are processed independently, allowing the same design to be applied without requiring aligned frames or matching camera viewpoints.

Figure S6 illustrates coordinated changes to the scene, clothing, and props across shots. The edited outputs maintain a recognizable shared setting and wardrobe while accommodating changes in body orientation, framing, and hand position. Facial appearance and recorded gestures remain recognizable as the surrounding visual design changes. These results demonstrate FlowCine's usefulness for coordinated restyling of recorded footage, with shared references providing visual guidance across independently edited shots.